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Open Problems

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Proper orientation number of r-partite graph

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This is joint work with Lanchao Wang and Xiaolin Wang

Let G be a simple undirected connected graph. A *proper orientation* D of G is an orientation of G such that $d_D^+(u) \neq d_D^+(v)$ for any edge $uv \in E(G)$. Denote the *proper orientation number* $\vec{\chi}(G)$ of G as the minimum $\Delta^+(D)$ among all proper orientations D of G, where $\Delta^+(D)$ is the maximum out-degree of D.

The *maximum average degree* of a graph G, denote as $\mathrm{Mad}(G)$, is the largest average degree over all subgraphs of G:

$$\mathrm{Mad}(G) = \max_{H \subseteq G} \frac{2|E(H)|}{|V(H)|}.$$

Chen, Mohar and Wu[1] proved that if G is a r-partite graph, then $\vec{\chi}(G) \leq \frac{1}{2}\mathrm{Mad}(G) + O(\frac{r \log r}{\log \log r})$. We recently construct some r-partite graphs with $\vec{\chi}(G) \geq \lceil \frac{1}{2}\mathrm{Mad}(G) \rceil + \lfloor \frac{5}{2} r \rfloor - 2$ and proved that $\vec{\chi}(G) \leq \lceil \frac{1}{2}\mathrm{Mad}(G) \rceil + 7$ for every 3-partite graph G.

Question 1. *Is this upper bound tight for 3-partite graphs?*

Question 2. *Can $\vec{\chi}(G) - \frac{1}{2}Mad(G)$ be bounded by a linear function of $\chi(G)$?*

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Ramsey–Turán Problems for Hypergraphs

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Let F be an r-uniform hypergraph. The Ramsey–Turán function $\mathrm{RT}(n, F, \ell)$ is defined as the maximum number of edges in an n-vertex F-free r-graph G with independence number less than ℓ, that is,

$$\mathrm{RT}(n, F, \ell) = \max\{e(G) : G \text{ is an } n\text{-vertex } F\text{-free } r\text{-graph and } \alpha(G) < \ell\}.$$

The corresponding Ramsey–Turán density is

$$\varrho(F) := \lim_{\delta\to 0} \lim_{n\to\infty} \frac{\mathrm{RT}(n, F, \delta n)}{\binom{n}{r}}.$$

A theorem of Erdős and Sós gives a large class of hypergraphs with Ramsey–Turán density zero.

Theorem 1. **[1]** *Let F be an r-graph with vertex set*

$$V(F) = \bigcup_{j=1}^{r} V_j$$

such that $F[V_r]$ is a linear forest, and all edges not contained in $F[V_r]$ are cross edges with respect to the partition $V_1, \dots, V_r$; that is, every such edge e satisfies

$$|e \cap V_j| = 1 \quad \text{for each } j \in [r].$$

Then $\varrho(F) = 0$.

The above theorem naturally leads to the question of what happens beyond this class. Let $F^{(3)}_{7,11}$ be the 3-graph with vertex set

$$\{x, y_1, y_2, y_3, z_1, z_2, z_3\}$$

and edge set

$$\{\{y_1, y_2, y_3\}, \{z_1, z_2, z_3\}, \{x, y_i, z_j\} : i, j \in [3]\}.$$

It seems to be one of the simplest examples not covered by Theorem 1.

Question 1. Determine the Ramsey–Turán density $\varrho\left(F^{(3)}_{7,11}\right)$.

More generally, one may ask whether Ramsey–Turán densities of 3-graphs exhibit a jump at zero.

Question 2. Is 0 a jump for Ramsey–Turán densities of 3-graphs? If yes, what is the smallest positive Ramsey–Turán density, and which hypergraphs attain it?

Another fundamental example is the Fano plane. Let $\mathbb{F}$ denote the Fano plane, namely the 3-graph with vertex set [7] and edge set

$$E(\mathbb{F}) = \{123, 147, 156, 246, 257, 345, 367\}.$$

Question 3. Determine the Ramsey–Turán density $\varrho(\mathbb{F})$.

Since the Fano plane is a linear 3-graph, it is also natural to consider the following broader problem.

Question 4. Can one determine, or at least give general upper and lower bounds for, the Ramsey–Turán densities of linear hypergraphs?

References

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The degree version for Erdős-Ko-Rado Theorem

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The famous Erdős-Ko-Rado Theorem states that if $\mathcal{F}$ is a k-uniform intersecting famliy, then $n \geq 2k$, $|\mathcal{F}| \leq \binom{n-1}{k-1}$. Moreover, equality holds when $|\mathcal{F}|$ is isomorphic to a 1-star, that is, the largest degree of $\mathcal{F}$ is $\binom{n-1}{k-1}$. For a set family $\mathcal{F} \subseteq 2^{[n]}$ and every element $i \in [n]$, let $d_i(\mathcal{F}) := |\{F \in \mathcal{F} : i \in F\}|$ denote the degree of i in $\mathcal{F}$. Without loss of generality, we assume that the degrees are ordered as $d_1 \geq d_2 \geq \cdots \geq d_n$. It is also an interesting question to determine the upper bound of the ℓ-largest degree of an intersecting family.

When $\ell \geq k+2$, 1-star may be the extremal case. Recently, Huang and Rao [1] proved that it holds when $\ell \geq 2k+1$ and $n \geq 12k, \ell \geq k+2$.

Question 1. *Can we find a constant M such that for every $n \geq 2k+M$, every intersecting family $\mathcal{F} \in 2^{[n]}$ satisfies*

$$d_{k+2} \leq \binom{n-2}{k-2}?$$

Remark. When $M = 2$ and $n = \{4, 5, 7\}$, there exist counterexamples. Let $k = 4, n = 10$. Construct $\mathcal{F}$ as the union of the following two families:

(a) all 4-element subsets of [6].

(b) all sets of the form $A \cup \{b\}$, where A is a block of a 2-$(6, 3, 2)$ design and $b \in \{7, 8, 9, 10\}$.

Nevertheless, when $\ell < k+2$, 1-star is no longer an extremal case.

Example (Hilton-Milner). Let $\mathcal{F} \subseteq \binom{[n]}{k}$ be the k-uniform intersecting set family consisting of the following subsets:

(a) all k-element sets containing the element 1 and at least one element from $2, 3, \ldots, \ell+1$;

(b) all k-element sets containing the set $2, 3, \ldots, \ell+1$.

It is not hard to check $d_{\ell+1}(\mathcal{F}) = \binom{n-2}{k-2} + \binom{n-\ell-1}{k-\ell}$. Frankl and Wang [1] prove that $d_{\ell+1}(\mathcal{F}$ when $n \geq 2\ell^2 k$, that is, $n = O(k)$ when ℓ is an constant. From another direction, Huang and Rao [1] prove the same inequality when $20 \leq \ell \leq k$, and $n \geq \frac{k^2}{\ell} + 64k$, that is, $n = O(k)$ when $\ell = \Omega(n)$. Therefore, Huang and Rao proposed the following question.

Question 2. *Can we find an absolute constant C such that for every $4 \leq \ell \leq k+1$ and $n \geq Ck$, any intersecting family $\mathcal{F} \in 2^{[n]}$ satisfies*

$$d_{\ell+1} \leq \binom{n-2}{k-2} + \binom{n-\ell-1}{k-\ell}?$$

If such a constant exists, what is the smallest possible value of C?

References

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Combinatorial Optimization in Microfluidic Biochip Routing

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This talk focuses on the cooperative multi-droplet routing problem in Digital Microfluidic Biochips (DMFBs). In DMFB synthesis, the primary objective of routing is to plan optimal paths for multiple droplets from source electrodes to target electrodes, while strictly satisfying static and dynamic fluidic constraints to avoid motion conflicts between droplets. This problem can be formulated as a constrained multi-path temporal planning problem on a grid graph, and has been proven to be NP-complete. However, as the scale of biochemical protocols expands, exact solution methods face exponential computational bottlenecks and are unable to meet the demands of real-time scheduling. How to achieve efficient real-time cooperative routing while ensuring constraint compliance has become an important research challenge and a key open frontier in this field.

The Erdős-Rogers Problem

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The Ramsey number $r_k(s,n)$ is the minimum N such that every red-blue coloring of the edges of $K_N^{(k)}$ contains a monochromatic red copy of $K_s^{(k)}$ or a monochromatic blue copy of $K_n^{(k)}$.

Definition 1. *For $2 \le k \le t < s$, the Erdős-Rogers function is*

$$f_{t,s}^{(k)}(N) = \min\{\alpha_t(G) : |V(G)| = N,\ G \text{ is } K_s^{(k)}\text{-free}\},$$

where $\alpha_t(G)$ is the largest size of a vertex set spanning no copy of $K_t^{(k)}$.

Remark.

- $f_{t',s}^{(k)}(N) \le f_{t,s}^{(k)}(N)$ if $t' \le t$ and $f_{t,s}^{(k)}(N) \le f_{t,s'}^{(k)}(N)$ if $s' \le s$.
- $r_k(s,n) = \min\{N : f_{k,s}^{(k)}(N) \ge n\}$.

In 2014, Dudek and Mubayi established that

$$\Omega\big((\log_{(k-2)} N)^{1/4+o(1)}\big) \le f_{t,t+1}^{(k)}(N) \le O\big((\log N)^{1/(k-2)}\big)$$

for each $t \ge k \ge 3$, where $\log_{(k-2)} N = \underbrace{\log\log\cdots\log}_{k-2} N$, and in the same year Conlon, Fox and Sudakov improved the lower bound by proving $f_{t,t+1}^{(k)}(N) \ge \Omega\big((\log_{(k-2)} N)^{1/3+o(1)}\big)$ for each $t \ge k \ge 3$.

In 2018, Mubayi and Suk[1] obtained $f_{k+1,k+2}^{(k)}(N) \le O(\log_{(k-13)} N)$ for $k \ge 14$ and put forward a conjecture claiming $f_{k+1,k+2}^{(k)}(N) = (\log_{(k-2)} N)^{\Theta(1)}$ for each $k \ge 4$, alongside a rephrased generalized conjecture:

Conjecture 1. $f_{k+1,s}^{(k)}(N) = (\log_{(k-2)} N)^{\Theta(1)}$ *for each $k \ge 4$ and $s \ge k+2$.*

Remark. Fan, Hu, Lin and Lu[2] gives $f_{k+1,s}^{(k)}(N) \le O\big(\log_{(k-3)} N\big)$ for each $k \ge 5$ and $s \ge k+2$, and D., Hu, Liu and Wang[3] showed that for every $s \ge 11$, $f_{5,s}^{(4)}(N) = (\log\log N)^{\Theta(1)}$, which yields the corresponding corollary $f_{k+1,s}^{(k)}(N) = \big(\log_{(k-2)} N\big)^{\Theta(1)}$ for each $k \ge 4$ and $s \ge k+7$.

Question 1. $f_{5,s}^{(4)} = (\log\log N)^{O(1)}$ *for $s \in [6,10]$?*

Question 2. $f_{5^-,5}^{(4)}(N) = (\log\log N)^{O(1)}$*?*

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Hamilton cycles in randomly sparsified Cayley graphs

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This talk is centered on Hamilton cycles in random subgraphs of Cayley graphs. A classical conjecture of Rapaport–Strasser asserts that every connected Cayley graph on a finite group with at least three elements is Hamiltonian. Recent work has verified this conjecture for moderately dense Cayley graphs.

The question I would like to discuss is the following random-sparsification version. Let G be a connected n-vertex Cayley graph of degree d, and let G_p be obtained by keeping each edge independently with probability p. Is there an absolute constant C such that G_p contains a Hamilton cycle with high probability whenever $pd \geq C \log n$?

This asks whether Cayley symmetry makes Hamiltonicity robust under random edge deletion down to logarithmic expected degree. I will also mention the current progress toward this question and the related weakening in which Hamilton cycles are replaced by perfect matchings.

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Ramsey numbers for sparse digraphs

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Let H be an acyclic digraph. The oriented Ramsey number of H, denoted by $\overrightarrow{r_1}(H)$, is the least integer N such that every tournament on N vertices contains a copy of H. Motivated by the classical Burr–Erdős Conjecture. Bucić, Letzter and Sudakovis[1] asked a natural directed analogue.

Problem. *Is it true that for every fixed Δ, every acyclic digraph H on n vertices with maximum degree at most Δ satisfies*

$$\overrightarrow{r_1}(H) = O_\Delta(n)?$$

Fox, He and Wigderson [2] showed that this is false in general. They proved that for every $\Delta \geq 2$ and every sufficiently large n,

$$n^{\Omega\left(\Delta^{2/3}/\log^{5/3}\Delta\right)} \leq \overrightarrow{r_1}(H) \leq n^{O_\Delta(\log n)}.$$

In particular, bounded maximum degree alone does not force a linear oriented Ramsey number.

On the other hand, linear bounds can still be obtained if the acyclic digraph has additional ordered structure. Bradač, Morawski, Sudakov and Wigderson [3] introduced the notion of graded bandwidth. They say that a digraph G has graded bandwidth at most w if its vertex set can be partitioned into

$$V_1, \ldots, V_H$$

such that for every edge $uv \in E(G)$ with $u \in V_i$ and $v \in V_j$, we have

$$1 \leq j - i \leq w.$$

They proved that if G has maximum degree Δ and graded bandwidth w, then

$$\vec{r}(G) \leq 3^{57\Delta w}|V(G)|.$$

Thus, the main difficulty is to understand which additional structural properties of sparse acyclic digraphs force a linear oriented Ramsey number.

Question 1. *For each fixed $\Delta \geq 2$, determine the growth rate of*

$$\max_{\substack{|V(H)|=n \\ H \text{ is acyclic} \\ \Delta(H)\leq\Delta}} \overrightarrow{r_1}(H).$$

In particular, is this maximum polynomial in n for every fixed Δ?

Question 2. *Which bounded-degree acyclic digraphs H satisfy*

$$\overrightarrow{r_1}(H) = O_\Delta(|V(H)|)?$$

Is there a natural ordered structural condition, beyond bounded maximum degree, which guarantees a linear oriented Ramsey number?

References

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The maximum size of cross-intersecting families and its relationship with intersecting family

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This talk is centered on the maximum sum of sizes of pairwise cross intersecting families.

A classical result of Erdős–Ko–Rado asserts that for $n > 2k$, the largest intersecting family of k-subsets of $[n]$ is a star.

Frankl and Tokushige later extended this to two families of different sizes.

Recent work by Huang, Peng determined the maximum sum for t families whose sizes are restricted to $k_1, k_2, \ldots, k_t$ respectively under the condition $n \geq k_1 + k_3$, and they claimed another paper has been prepared.

The question I would like to discuss is the intersecting version.

That means instead of considering t pairwise cross-intersecting families whose sizes are restricted to $k_1, k_2, \ldots, k_t$ respectively, we consider one intersecting family whose members' sizes could be chosen from the above $k_1, k_2, \ldots, k_t$.

We found that Huang Peng's results could be used in the corresponding situation, so we are wondering their following results under left restricted conditions of n.

References

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Clique Decomposition Thresholds and Fractional Obstructions Beyond Nash–Williams

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A K_q-decomposition of a graph G is a partition of $E(G)$ into copies of K_q. The natural divisibility conditions are

$$\binom{q}{2} \mid e(G) \quad \text{and} \quad q-1 \mid d_G(v), \quad \forall v \in V(G).$$

Let δ_{K_q} be the asymptotic minimum degree threshold for guaranteeing a K_q-decomposition in every sufficiently large K_q-divisible graph. The classical Nash–Williams conjecture says

$$\delta_{K_3} = \frac{3}{4}.$$

A natural folklore extension predicted $\delta_{K_q} = 1 - \frac{1}{q+1}$ for all $q \geq 4$, but this has recently been disproved by graphs with no fractional K_q-decomposition [4]. Since for cliques the exact threshold equals the fractional threshold, namely

$$\delta_{K_q} = \delta^*_{K_q},$$

the problem is essentially reduced to understanding fractional obstructions [3].

Question. *Is $\delta_{K_3} = 3/4$? For $q \geq 4$, determine*

$$\delta_{K_q} = \delta^*_{K_q}.$$

Equivalently, among all sufficiently large graphs with no fractional K_q-decomposition, what is the largest possible asymptotic minimum degree?

This talk will discuss these above threshold questions.

References

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Bandwidth theorem in digraphs

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This talk is centered on a possible bandwidth theorem for digraphs. I will begin with the undirected setting, recalling the classical bandwidth theorem in graphs [1]. I will then move to the directed setting and explain some of these new difficulties through examples These examples show that the directed setting is not just a simple copy of the undirected one.

The goal of the talk is to explain how the idea of bandwidth may be moved from graphs to digraphs, what new barriers appear in this process, and what kind of structure a digraph bandwidth theorem should describe.

References

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Cycle decomposition of Eulerian digraph

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The problem. Bollobás and Scott (1996) conjectured that every n-vertex Eulerian digraph can be decomposed into $O(n)$ edge-disjoint directed cycles. This is the directed analogue of the Hajós conjecture (still open for undirected graphs).

Current best bound for digraphs. Knierim, Larcher, Martinsson and Noever (2021) proved that any Eulerian digraph with maximum outdegree Δ admits a cycle decomposition using at most $O(n \log \Delta)$ cycles.

What is known for undirected graphs? For undirected graphs, the current record is $O(n \log^* n)$ cycles and edges, due to Bucić and Montgomery (2023) [1], improving upon a long line of results ($O(n \log \log n)$, etc.).

Our attempt and the obstacles. We try to import the expander-based method from the undirected setting to directed graphs, hoping to replace $O(n \log \Delta)$ by $O(n \log^* n)$ (or even $O(n)$). However, several difficulties arise:

- Directed expander decomposition is less flexible; preserving Eulerian property after removing cycles is nontrivial.

We will discuss these challenges and invite ideas on possible modifications (e.g., using weighted auxiliary graphs, or a hybrid of the weighting and expander methods).

Open questions for discussion.

1. Can the expander decomposition be adapted to directed Eulerian graphs to achieve $O(n \log^* n)$ cycles?

References

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The exact value of $inv(Q_n)$

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For an oriented graph D and a set $X \subseteq V(D)$, the inversion of X in D is the graph obtained by reversing the orientation of every arc with both ends in X. The inversion of a collection $X_1, X_2, \ldots, X_m$ is the graph obtained by successively inverting each set. Note that the order in which this is done does not matter the result graph. $X_1, X_2, \ldots, X_m$ is called a decycling family for D if its inversion contains no directed cycle.

The inversion number $inv(D)$ is defined by Belkhechine in [1], it denotes the minimal integer m such that there exists an m-decycling family $X_1, X_2, \ldots, X_m$ of D. Some initial conclusions of the inversion number were proved by Belkechine, Bouaziz, Boudabbous, and Pouzet [2].

Definition (Inversion Number). *A collection $X_1, X_2, \ldots, X_m$ is called a decycling family for D if its inversion contains no directed cycles. The inversion number, denoted by $inv(D)$, is the minimal integer m such that there exists an m-decycling family for D.*

Definition (The Tournament Q_n). *The tournament Q_n is defined on the vertex set $[n]$ with the arc set $A(Q_n) = \{ij \mid i, j \in [n], j > i + 1\} \cup \{ji \mid i, j \in [n], j = i + 1\}$.*

Note that, equivalently, Q_n can be obtained by reversing a Hamilton path of a transitive tournament on n vertices.

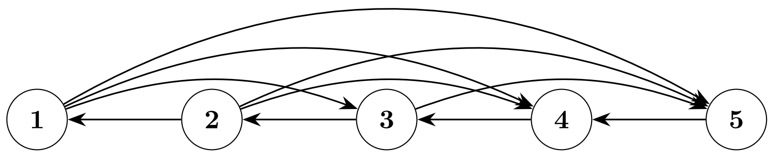

Figure 1. Tournament Q_5

Conjecture[3]. *For the tournament Q_n on n vertices, the inversion number is given by:*

$$inv(Q_n) = \left\lfloor \frac{n-1}{2} \right\rfloor$$

.

The equation has been verified for $3 \leq n \leq 7$ [4], but remains open for larger n.

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